

Yeeun Lee

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RESEARCH INTERESTS

Machine-Learning Interatomic Potentials, Multiscale Molecular Simulation, Molecular Coarse-Graining, Energy Materials, Solid State Electrolytes, Electrochemical Interfaces.

RESEARCH EXPERIENCE

Graduate Research

Apr. 2024 - Present

1) Technische Universität München (TUM), Germany

Advisor: Prof. Patrick Rinke

- **Project: Machine-Learning Molecular Dynamics of Electric-Field-Induced Vibrational Dynamics**

- Performed molecular dynamics simulations under external electric fields using pretrained MACE machine-learning interatomic potentials in Linux/Slurm HPC environments.
- Analyzed MD trajectories using Python to extract time-dependent dipole responses, IR spectra, and mode-resolved vibrational energies under different electric-field conditions.
- Investigated field-induced vibrational behavior and energy transfer and benchmarked ML-MD results against DFT reference calculations and experimental observations.

2) Forschungszentrum Jülich, Germany

Advisor: Dr. Baohu Wu

- **Project: Operando Investigation of Structural and Interfacial Evolution in PVDF-HFP-Based Polymer Electrolytes for Li Solid-State Batteries**

- Synthesized PVDF-HFP/Succinonitrile-based polymer electrolytes and prepared operando-cell configurations
- Combined in situ Raman spectroscopy, SAXS/WAXS, operando battery cycling, and electrochemical testing to correlate structural evolution with electrochemical behavior.

3) Technische Universität München (TUM), Germany

Advisor: Prof. Jennifer L. M. Rupp

- **Project 1: Electrochemical and Thermal Analysis of Prussian Blue Analogue (PBA) Thin Films for Thermobatteries**

- Investigated the electrochemical and thermal behavior of PBA thin films for thermobattery applications.

- **Project 2: WO₃ Thin Films via Spray Pyrolysis for Proton-Based Energy Storage**

- Synthesized WO₃ thin films by spray pyrolysis while systematically varying substrate temperature and spray volume to control film morphology and thickness.
- Correlated SEM, Raman spectroscopy, XRD, cyclic voltammetry, and EIS results to establish processing–structure–electrochemical property relationships.
- Identified strong processing-dependent variations in film thickness, proton intercalation, and charge-transfer resistance, and determined deposition conditions that promoted reversible proton intercalation/deintercalation and stable electrochemical performance.

Undergraduate Research

Mar. 2023 - Sep 2023

1) Korea University, Seoul, Korea

Advisor: Prof. Seok Su Sohn

- **Project: Hydrogen embrittlement behavior in CoCrFeMnNi high-entropy alloys (HEAs) under cryogenic condition**

Research Intern

May. 2022 - Jan. 2023

1) SK On, Joint Battery Platform Project with Korea University

- **Project: Smart Battery & EV Charging Platform Development**

EDUCATION

M.S. in Materials Science & Engineering **Sep. 2026**

Technical University of Munich (TUM), (Munich, Germany)

Overall GPA: 1.6 (German grading scale; 1.0 = best; 5.0 = fail)

B.S. in Materials Science & Engineering **Feb. 2024**

Korea University, (Seoul, Korea)

PUBLICATION

†(co-)first author, *(co-)corresponding author

1. Y. Lee, B. Wu*. Operando Investigation of Structural and Interfacial Evolution in PVDF-HFP/Succinonitrile Polymer Electrolytes. (*In preparation*)
2. Y. Lee†, N. Isaev†, J. L. M. Rupp*. Proton Intercalation in WO₃ Thin Films Synthesized by Spray Pyrolysis under Varying Parameters. (*In preparation*)
3. P. Borah†, Y. Lee, A. Bharati*. Small-Angle Neutron Scattering Analysis of Pectin Conformation in Solution. (*In preparation*)
4. N. Bhatia†, Y. Lee†, P. Rinke*. Machine-Learning Molecular Dynamics of Electric-Field-Induced Vibrational Dynamics. (*In preparation*)

PRESENTATIONS

1. N. Isaev, Y. Lee, K. Bohnet, J. L. M. Rupp*. “Tunable Proton Intercalation in Tungsten Oxide Thin Films for Proton-Based Energy Storage Devices.” E-MRS 2026 Spring Meeting, Strasbourg, France, May 25-29, 2026.
2. Y. Lee, A. Bharati, B. Wu*. “Operando Structural Evolution of PVDF-HFP Polymer Electrolytes in Lithium Batteries.” 7th International Energy Storage Materials Conference, Dresden, Germany, Sep 20-23, 2026.
3. Y. Lee, A. Bharati, B. Wu*. “Operando Structural Evolution of PVDF-HFP Polymer Electrolytes in Lithium Batteries.” MLZ User Meeting 2026, Munich, Germany. Dec 2-3, 2026.

SCHOLARSHIPS

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- **Merit-based Waiver Scholarship for Academic Excellence** **Oct. 2025 - Sep. 2026**
Technical University of Munich (TUM)
 - Awarded to the top-performing student in the department based on academic excellence

HONORS & AWARDS

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- **Capstone design Award**, Korea university **Sep. 2023**
 - **Excellence Award**, SK-On Battery Platform Research **Feb. 2023**
 - **The Grand Award**, AI model for predicting solar panel power generation contest. **July. 2023**
 - **Outstanding Organic Chemistry Mentoring Award**, korea university **Mar. 2022**

EXTRACURRICULAR ACTIVITIES

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- **AI Solar Power Prediction Contest**, Korea University **Feb. 2023 - July. 2023**
 - Developed a stacking ensemble for solar power forecasting, integrating tree-based and deep-learning models including LightGBM, Random Forest, MLP, and LSTM-based architectures.
 - Evaluated and optimized predictive performance for time-series energy forecasting, achieving 1st Place in the competition.

VOLUNTEER ACTIVITIES

- **Community Support Program for the Visually Impaired** **Mar. 2022 - Aug. 2023**
 - Assisted visually impaired individuals through community service activities, promoting accessibility and inclusion.
- **Educational Mentoring Program for Underprivileged Students** **Nov. 2021 - Feb. 2022**
 - Provided academic mentoring and educational support for students in marginalized communities, aiming to promote equal learning opportunities and address educational disparities.

SKILLS

- **Computational/Programming**
 - Python, Linux, Slurm, Matlab, Machine Learning, Molecular dynamics
- **Electrochemical & Battery Testing**
 - Battery cell assembly, Operando battery cycling, BioLogic EC-Lab
- **Materials Characterization**
 - In-Situ Raman, SAXS/WAXS, XRD, AFM, SEM, DSC, TGA, FTIR, EBSD, Tensile testing
- **Materials Processing & Synthesis**
 - Tape casting, Solution casting, Spray pyrolysis, Electrochemical deposition, Glovebox handling